Coordinate-free Mathematics

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1 Linear Algebra

Linear algebra studies linear maps between vector spaces. A **vector space** V over a field k is a set equipped with vector addition and scalar multiplication operations satisfying straightforward compatibility conditions. A **linear map** $f\colon V \to U$ between k-vector spaces is a map of underlying sets such that $f(\alpha v + \beta w) = \alpha f(v) + \beta f(w)$ for all $\alpha, \beta \in k$ and $v, w \in V$. Examples include differential operators, group representations, coefficient matrices of linear systems of equations, and local approximations of globally nonlinear functions.

The **free k-vector space on a set** X, denoted kX, is the set of finitely supported set-theoretical functions $X \to k$ with pointwise addition and scalar multiplication. The **direct sum** vector space $V \oplus W$ is the set $V \times W$ with pointwise addition and scalar multiplication defined by $\alpha(v \oplus w) = \alpha v \oplus \alpha w$. The **quotient** of V by a subspace W, denoted V/W, is the set of equivalence classes of V modulo the relation $x \sim y$ iff $x - y \in W$. The **tensor product** $V \otimes W$ is the vector space $k(V \times W)/N$ where N is the subspace generated by the linearity relations on V and W.

Exercise 1 *An* ***isomorphism*** *of vector spaces $V \cong W$ consists of a pair of linear maps $f\colon V \to W$ and $g\colon W \to V$ such that $f \circ g = 1_W$ and $g \circ f = 1_V$. Prove the following:*

- $k(X \sqcup Y) \cong kX \oplus kY$.
- $k(X \times Y) \cong kX \otimes kY$.
- *The first isomorphism theorem for vector spaces (and the second and third if desired).*
- *Find universal properties determining up to canonical isomorphism the free k-vector space, direct sum, quotient, tensor product, and dual space (defined below), respectively.*

The **dual space** V^* is the space of linear maps from a vector space V to its ground field k. The evaluation map $\mathrm{ev}_V\colon V^* \otimes V \to k$ is the linear extension of $f \otimes x \mapsto f(x)$. A linear map $T\colon V \to V$ induces the dual map $T^*\colon V^* \to V^*$ defined by $T^*(\phi) = \phi \circ T$.

Exercise 2 *Let $T\colon V \to V$ describe the evolution of a linear system. Prove the following:*

- *A linear observable $\phi \in V^*$ is conserved iff $T^*\phi = \phi$.*
- *The canonical map $V \to V^{**}$ sends a state to the rule assigning to each observable its value on that state.*
- *A state is determined by the values of all linear observables when V is finite dimensional.*

The tensor product introduced above is the state space of a composite linear system.

Exercise 3 *Let V and W be state spaces of two systems. Prove the following:*

- *Bilinear observables on $V \times W$ correspond canonically to linear observables on $V \otimes W$.*
- *Not every element of $V \otimes W$ is of the form $v \otimes w$ when $\dim(V), \dim(W) > 1$.*
- *For $\psi \in V \otimes W$, contraction defines a map $\widetilde{\psi}\colon V^* \to W$, and ψ is decomposable iff $\widetilde{\psi}$ has rank at most one.*

Non-decomposable tensors represent entangled states.

We say that V is **finite dimensional** if there is a coevaluation morphism $\mathrm{coev}_V\colon k \to V \otimes V^*$ satisfying the snake equations

$$(1_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes 1_V) = 1_V, \qquad (\mathrm{ev}_V \otimes 1_{V^*}) \circ (1_{V^*} \otimes \mathrm{coev}_V) = 1_{V^*}.$$

For $f \in \mathrm{End}(V)$, define

$$\mathrm{tr}(f) = (\mathrm{ev}_V \circ s_{V,V^*} \circ (f \otimes 1_{V^*}) \circ \mathrm{coev}_V)(1),$$

where $s_{X,Y}\colon X \otimes Y \to Y \otimes X$ is the symmetric braiding $x \otimes y \mapsto y \otimes x$. Define $\dim(V) = \mathrm{tr}(1_V)$.

Exercise 4 *Let V and W be finite dimensional. For $F \in \mathrm{End}(V \otimes W)$, use evaluation, coevaluation, and the symmetric braiding to define*

$$\mathrm{tr}_W(F) = (1_V \otimes \mathrm{ev}_W) \circ (1_V \otimes s_{W,W^*}) \circ (F \otimes 1_{W^*}) \circ (1_V \otimes \mathrm{coev}_W).$$

Prove that this gives a map $\mathrm{tr}_W\colon \mathrm{End}(V \otimes W) \to \mathrm{End}(V)$, that it is natural under isomorphisms of V and W, and that

$$\mathrm{tr}_W(f \otimes g) = \mathrm{tr}(g) f.$$

Exercise 5 *Prove the following for finite-dimensional vector spaces V, W, endomorphisms $f, g \in \mathrm{End}(V)$, and $h \in \mathrm{End}(W)$:*

- $\operatorname{tr}(fg) = \operatorname{tr}(gf)$.

- $\operatorname{tr}(f \oplus h) = \operatorname{tr}(f) + \operatorname{tr}(h)$.

- $\operatorname{tr}(f \otimes h) = \operatorname{tr}(f)\operatorname{tr}(h)$.

- $\dim(V) \in \mathbb{N}$.

- $V \cong W$ *iff* $\dim(V) = \dim(W)$.

Suppose $k = \mathbb{C}$. A Hermitian inner product $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{C}$ is a positive-definite sesquilinear form, linear in the first variable, conjugate linear in the second variable, and conjugate symmetric.

Exercise 6 *For a nonzero vector ψ in a finite-dimensional Hermitian vector space V, define*

$$p_\psi(v) = \frac{\langle v, \psi \rangle}{\langle \psi, \psi \rangle} \psi.$$

Prove that p_ψ is a self-adjoint idempotent of rank one and depends only on the one-dimensional subspace spanned by ψ. Conclude that a pure state is naturally a ray rather than a distinguished vector.

A linear map $f \colon V \to V$ is **self-adjoint** if $\langle fv, w \rangle = \langle v, fw \rangle$ for all $v, w \in V$.

Exercise 7 *Prove that any self-adjoint linear map f on a finite-dimensional complex vector space V is diagonalizable:*

$$f = \sum_i \lambda_i p_i,$$

where $\sum_i p_i = 1_V$ and $p_i p_j = \delta_{ij} p_i$. Use this fact to prove the existence of the singular value decomposition for an arbitrary linear map between finite-dimensional complex vector spaces.

A linear map $U \colon V \to V$ is **unitary** if $U^{-1} = U^*$, where the adjoint is determined by $\langle fv, w \rangle = \langle v, f^* w \rangle$.

Exercise 8 *Let $U \colon V \to V$ be unitary. Prove the following:*

- *U preserves inner products, orthogonality, and rank-one projections.*

- *Conjugation $a \mapsto U^{-1} a U$ preserves self-adjointness, sums, products, and spectra.*

- *The transformations of states by U and of observables by conjugation give equivalent descriptions of the same evolution.*

A scalar $\lambda \in k$ is an **eigenvalue** for a linear map $f\colon V \to V$ if there exists a nonzero vector $v \in V$ for which $fv = \lambda v$. Its **eigenspace** is

$$\ker(f - \lambda 1_V),$$

whose nonzero elements are eigenvectors; its dimension is the **geometric multiplicity** of λ.

Exercise 9 *Let a and b be self-adjoint endomorphisms of a finite-dimensional Hermitian vector space. Prove that if $ab = ba$, then every eigenspace of a is preserved by b. Deduce that*

$$V = \bigoplus_{\lambda,\mu} V_{\lambda,\mu}$$

as an orthogonal direct sum of simultaneous eigenspaces.

Exercise 10 *Let a group G act linearly on V, and let $h \in \mathrm{End}(V)$ commute with this action. Prove that every eigenspace of h is a representation of G. Explain why an irreducible representation of dimension greater than one contained in an eigenspace forces the corresponding eigenvalue to be degenerate.*

The **tensor algebra** and **exterior algebra** of V are

$$T(V) = \bigoplus_{k\geq 0} V^{\otimes k}, \qquad \Lambda V = T(V)/\langle v \otimes v : v \in V\rangle.$$

Define $\Lambda^k V$ to be the degree-k component of ΛV.

Exercise 11 *If V is finite dimensional, show that the highest nonzero graded component of ΛV is one-dimensional.*

The **determinant** $\det(f)$ is the coefficient of the linear map on the highest-graded component of ΛV given by

$$v_1 \wedge \cdots \wedge v_{\dim(V)} \longmapsto f(v_1) \wedge \cdots \wedge f(v_{\dim(V)}).$$

Exercise 12 *Prove the following:*

- $\det(f \oplus g) = \det(f)\det(g)$.
- *If $V = \mathbb{R}^n$, then $\det(f)$ is the signed volume of $f(I^n)$ where $I = [0,1]$.*
- *Cramer's rule.*

For $f \in \mathrm{End}(V)$, the **characteristic polynomial** is

$$\lambda \longmapsto \det(f - \lambda 1_V).$$

Exercise 13 *Let f be a linear operator on a finite-dimensional vector space V. Assume that its characteristic polynomial splits over k. Prove the following:*

- *Every eigenvalue of f is a root of its characteristic polynomial.*
- *The geometric multiplicity of each eigenvalue is at most its **algebraic multiplicity**, its multiplicity as a root of the characteristic polynomial.*
- $\operatorname{tr}(f) = \sum_i \lambda_i$ *and* $\det(f) = \prod_i \lambda_i$*, where each eigenvalue λ_i is repeated according to its algebraic multiplicity.*
- *The operator f satisfies its own characteristic polynomial.*
- *When $k = \mathbb{C}$, the operator f has a Jordan normal form.*

2 Calculus

A **smooth manifold** is a topological space that is locally diffeomorphic to Euclidean space $\mathbb{R}^n$. A **smooth vector bundle** $p\colon E \to M$ consists of a smooth surjection p between smooth manifolds E and M such that all fibers are isomorphic to a single vector space and smooth local trivializations exist. A prototypical example is the **tangent bundle** TM, which consists of all tangent spaces of points of a manifold. A **vector field** $X \in \Gamma(TM)$ is a smooth section of the tangent bundle.

Exercise 14 *Write down precise intrinsic definitions for the following:*

- *Smooth manifold, smooth map, vector bundle, and tangent bundle.*
- *Bundle map, direct sum, tensor product, and dual bundle.*
- *Derivative of a smooth function between manifolds.*

For a configuration manifold M, velocities are tangent vectors and linear measurements of velocities are cotangent vectors. An inner product may identify these spaces, but this identification is additional structure.

Exercise 15 *Let $p \in M$. Prove the following:*

- *A velocity at p is naturally an element of T_pM, whereas momentum and force are naturally elements of T_p^*M.*
- *The pairing $\alpha(v)$ for $\alpha \in T_p^*M$ and $v \in T_pM$ is invariant under changes of coordinates.*
- *An inner product on T_pM induces an isomorphism $T_pM \cong T_p^*M$, but no such isomorphism is canonical without additional structure.*

Interpret the scalar $\alpha(v)$ as infinitesimal work when α is force and v is displacement.

To define integration in a coordinate-free way, we integrate differential forms over chains. Write $\Omega^n(M) = \Gamma(\Lambda^n T^*M)$ for the space of **differential n-forms**. The **exterior derivative** is the natural graded derivation $d\colon \Omega^n(M) \to \Omega^{n+1}(M)$ extending the differential of smooth functions and satisfying $d^2 = 0$.

The de Rham cohomology of M is

$$H^n_{\mathrm{dR}}(M) = \ker(d\colon \Omega^n(M) \to \Omega^{n+1}(M))/\operatorname{im}(d\colon \Omega^{n-1}(M) \to \Omega^n(M)).$$

The equation $dF = 0$ is the coordinate-free homogeneous Maxwell equation.

Exercise 16 *Let M be a four-dimensional spacetime and let $F \in \Omega^2(M)$. Prove the following:*

- *F assigns a scalar flux to each oriented infinitesimal surface.*
- *If $F = dA$ for some $A \in \Omega^1(M)$, then replacing A by $A + d\lambda$ does not change F.*
- *A global potential A exists for a closed form F iff $[F] = 0$ in $H^2_{\mathrm{dR}}(M)$.*

A **singular n-simplex** is a smooth map $\sigma\colon \Delta^n \to M$. Singular n-simplices freely generate the abelian group $C_n(M)$ of **singular n-chains**. The **boundary map** $\partial\colon C_n(M) \to C_{n-1}(M)$ is defined by

$$\partial\sigma = \sum_{j=0}^{n} (-1)^j \sigma \circ f_j,$$

where $f_j\colon \Delta^{n-1} \to \Delta^n$ includes the face opposite the j-th vertex.

A **chain complex** $(A_n, d_n)_n$ is a sequence of abelian groups and homomorphisms $d_n\colon A_n \to A_{n-1}$ such that $d_{n-1} \circ d_n = 0$. Its n-th **homology group** is

$$H_n = \ker(d_n)/\operatorname{im}(d_{n+1}).$$

Singular homology is the homology of the singular chain complex. Cohomology and cochains are defined similarly, with coboundary maps increasing degree; singular cohomology is obtained by dualizing the singular chain complex.

Exercise 17 *Prove the following:*

- *Differential forms on M form a cochain complex under the exterior derivative.*
- *Define the integral of an n-form on an n-chain and prove $\int_c d\omega = \int_{\partial c} \omega$ for every n-chain c and differential $(n-1)$-form ω.*
- *Singular cohomology with real coefficients and de Rham cohomology are naturally isomorphic.*

- *Bounded chain complexes of finite-dimensional vector spaces form a symmetric monoidal category with duals; the trace of an endomorphism is its Lefschetz number and the dimension of a complex is its Euler characteristic.*

Exercise 18 *Interpret*

$$\int_{\partial c} \omega = \int_c d\omega$$

as a relation between a quantity measured on a boundary and its source in the interior. Give conceptual proofs that the fundamental theorem of calculus, circulation–curl, flux–divergence, and Faraday induction are instances of this identity after the relevant geometric structures are chosen.

An **affine connection** is a map $\nabla\colon \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$ such that $\nabla_{fX} Y = f\nabla_X Y$ and $\nabla_X(fY) = X(f)Y + f\nabla_X Y$ for every smooth scalar field f on M.

A symplectic manifold is a smooth manifold M equipped with a closed non-degenerate two-form ω. For $H \in C^\infty(M)$, the Hamiltonian vector field X_H is determined by

$$\iota_{X_H}\omega = dH.$$

Exercise 19 *Suppose* $\dim(M) = 2n$. *Prove the following:*

- $\omega^n/n!$ *is a volume form.*
- *Every diffeomorphism preserving* ω *preserves this volume form.*
- H *is constant along the flow of* X_H.
- *An observable* $f \in C^\infty(M)$ *is conserved iff* $df(X_H) = 0$.

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